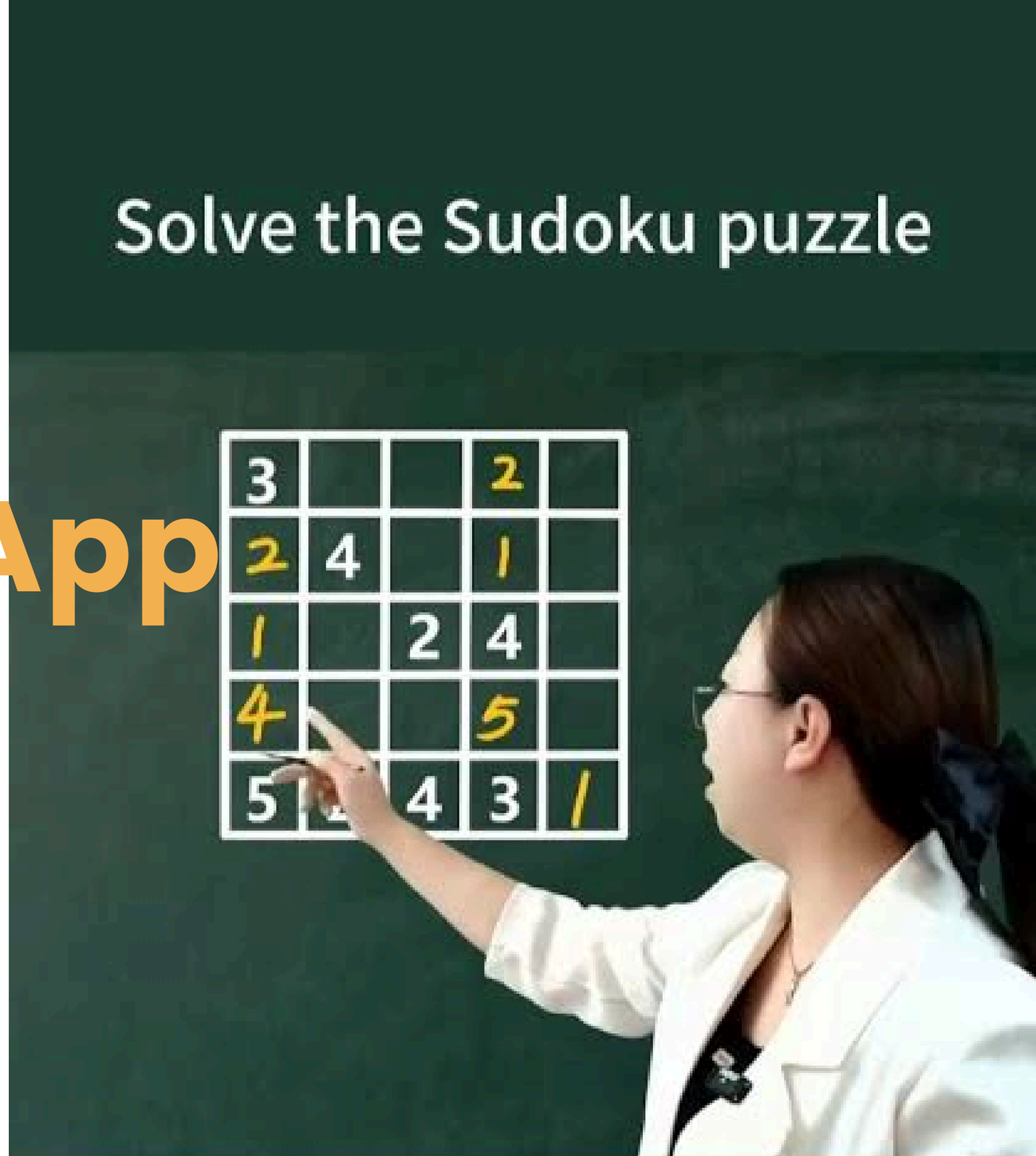
CADT
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Institute of Digital Technology

QUANTUM COMPUTING · PROJECT PROPOSAL

QUSUD Web App

Solving Sudoku with a quantum search algorithm



Team: Quantum Lover
Gen10 G2 · Software Engineering
Lecturer: Sovanmonynuth Heng

MEET OUR TEAM



Choeng Rayu



Tep Samnang



Sophal Taingchhay



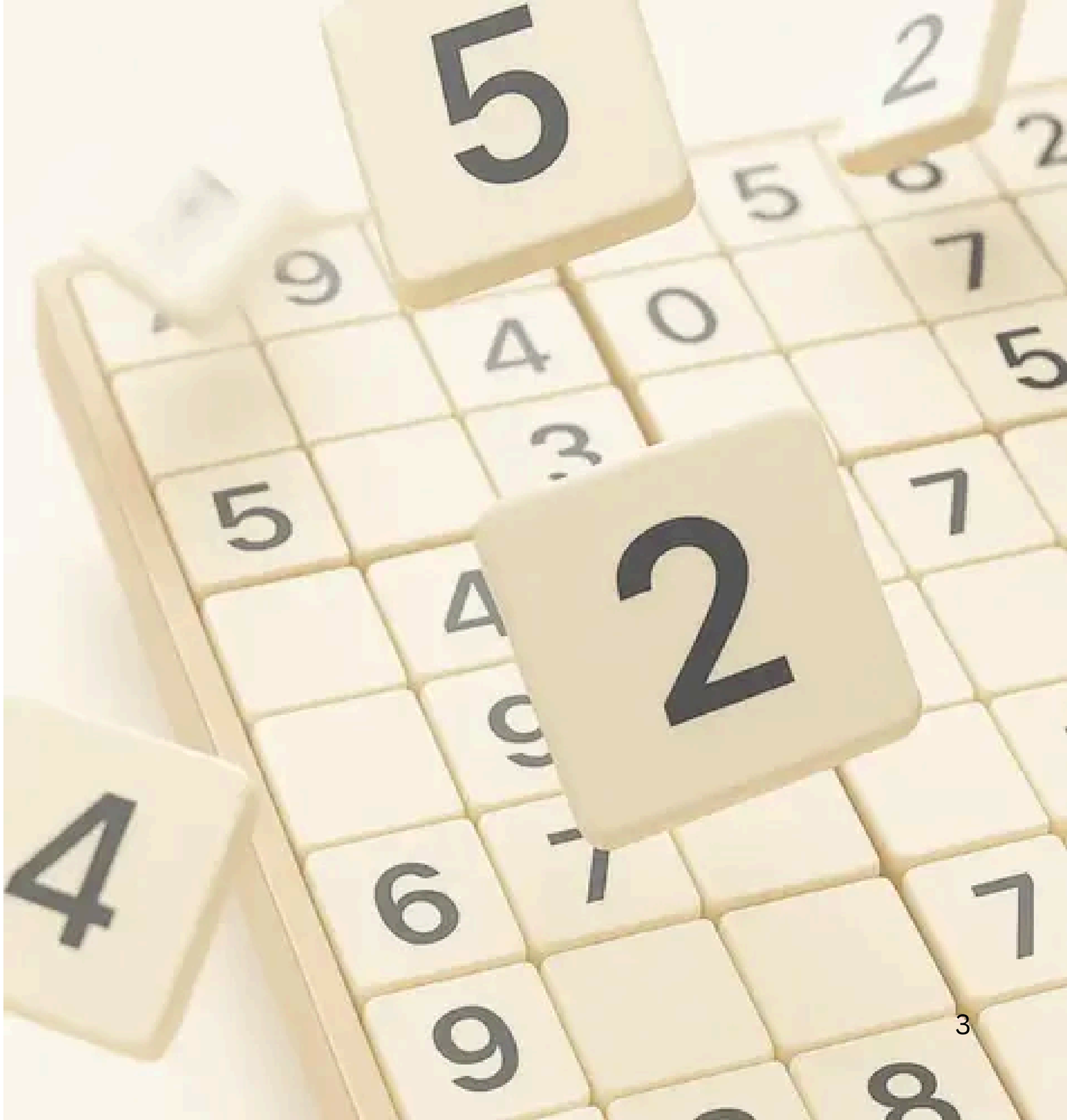
Ratana Aslnike



Um Lyrithyreach

CONTENT

1. Introduction
2. Objectives & scope
3. Algorithm & method
4. System & technologies
5. Plan



Introduction

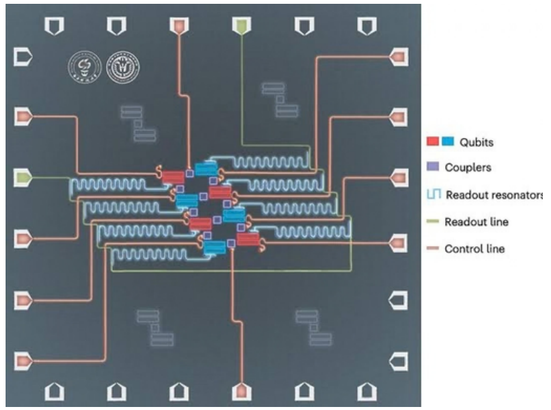
Introduction

QuSud is an Web app using quantum program finds the missing numbers.

5	3	9				4		
7	2	8	3		4	9		
6	4	1				7	3	
4	6	2	5	3	9	8	7	1
3	8	5	7	2	1	6	4	9
1	9	7	4	6	8	2	5	3
2	5	6	1	8	7	3	9	4
9	1	3		4		5	8	7
8	7	4	9	5	3	1	2	6

1. Take a photo

A 4×4 Sudoku with empty cells



2. Quantum search

Checks all possible answers together

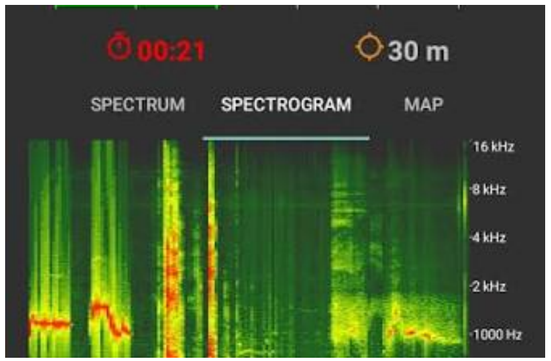
1	**2**	3	4
3	4	**1**	2
2	1	4	3
4	3	2	**1**

3. Get the answer

The missing numbers are filled in

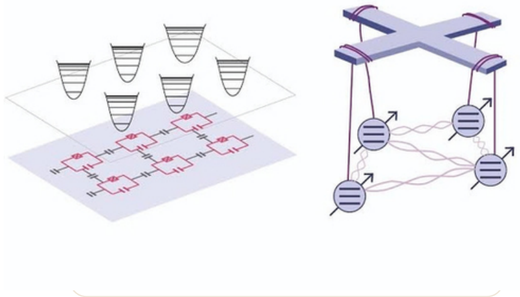
PROBLEM STATEMENT

Quantum computers make mistakes.



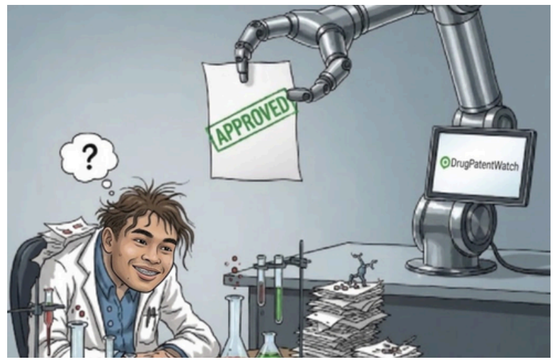
Noise

Tiny errors happen at every step on the chip



Ignored

Most demos use only a perfect simulator



Unclear

Hard to see quantum search on a real problem

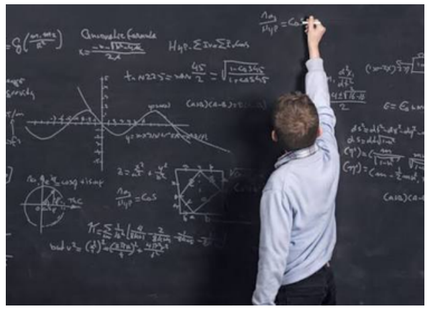
☹ **A noisy quantum answer**

1	**3**	3	4
3	4	**4**	2
1	1	4	3
4	3	2	**4**

Oops: two 3s in the top row!

PROPOSED SOLUTION

QuSud: solve it, test it, fix it.



1 · Solve

Grover’s search finds the missing numbers



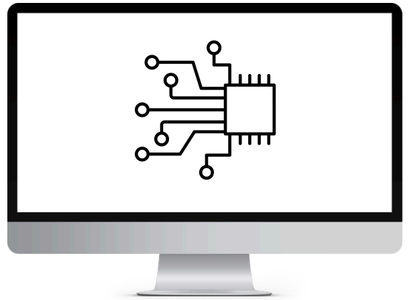
2 · Test

Run it on a noisy Simulator



3 · Fix

Mitiq cleans up the mistakes



4 · Share

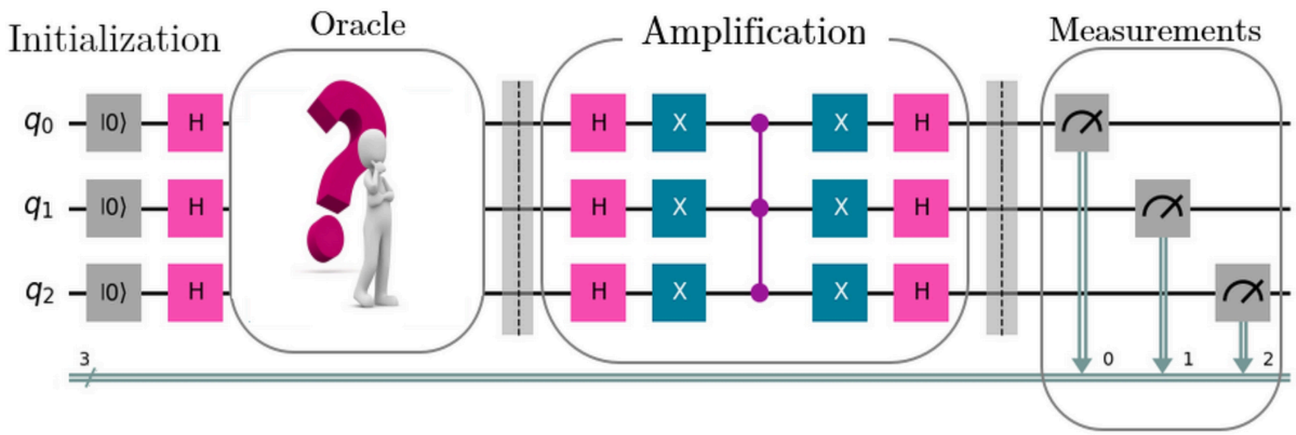
Photo in, answer out, in a web app

A **QuSud** app that uses a quantum program and cleans up its mistakes.

Objectives & scope

Project Objective

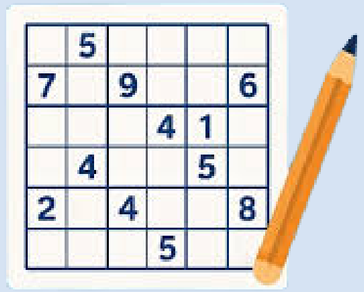
To explore how quantum computing, using Grover's Search Algorithm, can be applied to search for valid solutions to a 4×4 Sudoku puzzle.



1	2	3	4
3	4	2	1
2	1	4	3
4	3	1	2

Specific Objective

1



Develop an **web-app** that allows users to upload a photo of a 4×4 Sudoku puzzle

2

OpenCV

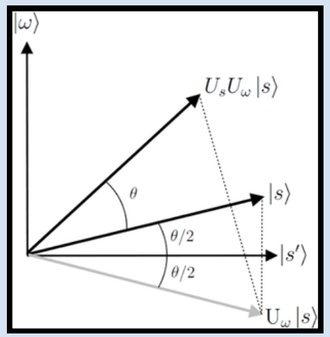
Use **OpenCV** to recognize and extract numbers from the uploaded

3

Convert Sudoku rules and constraints into a quantum circuit

Specific Objective

4



Apply **Grover's Search** Algorithm to search

5



Evaluate the algorithm under ideal and noisy quantum simulation

6

mitiq

Apply quantum error mitigation using **Mitiq**

PROJECT SCOPE

In scope our project will:

- Support 4×4 Sudoku puzzles with empty cells (we will scale it soon)
- Upload Sudoku images as input
- Use **OpenCV** for number recognition and puzzle extraction
- Implement **Grover's Search Algorithm**
- Simulate quantum circuits
- Display the recognized and solved Sudoku puzzle

Out of scope:

- Supporting standard 9×9 Sudoku puzzles
- Recognizing handwritten
- Running experiments on real quantum hardware
- Developing advanced quantum error correction
- Guaranteeing quantum speed

Algorithm & method

HOW GROVER'S SEARCH WORKS

Why Grover? Solving a Sudoku means searching for the one answer that follows all the rules, and Grover is the quantum algorithm for searching.

How it works, like finding **1 right key** among 256:



1 · Try all

Every answer starts equal (superposition)

2 · Mark

The oracle flips the right answer

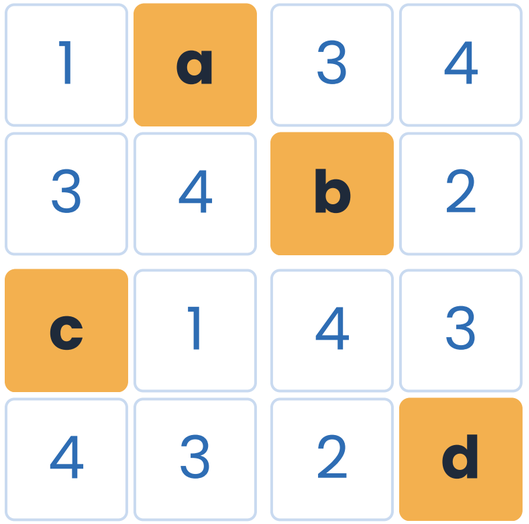
3 · Boost

The diffuser makes it bigger, others smaller

4 · Look

Measure: the right answer appears most

PUTTING SUDOKU INTO QUBITS



STEP 1 · 2 QUBITS = 1 NUMBER

1 = 00 **2 = 01** **3 = 10** **4 = 11**

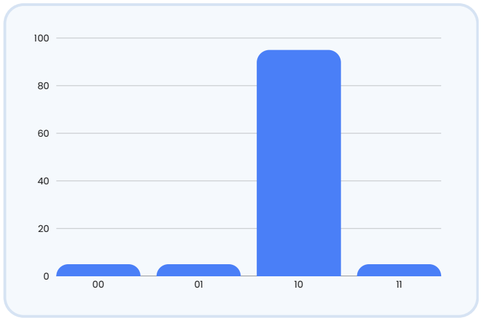
STEP 2 · COUNT

4 empty cells × 2 qubits = **8 qubits = 256 possible answers**

STEP 3 · CHECK THE RULES

No number repeats in a row, column or box.

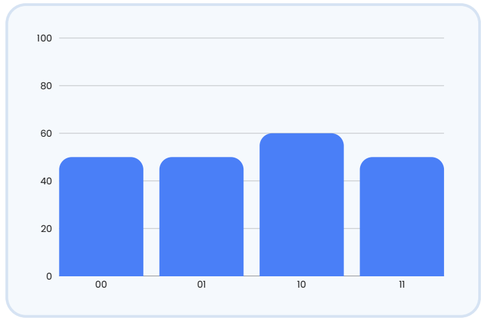
QUANTUM COMPUTERS MAKE MISTAKES



Perfect simulator

No mistakes: the right answer wins clearly

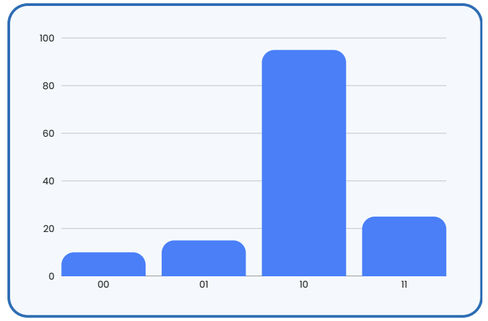
Right answer



Noisy computer

Tiny errors (noise) mix up the result

Often wrong



Noisy + error fixing

Error mitigation (Mitiq) cleans it up

Right again

We test with **FakeKolkata**: a free copy of a real 27-qubit IBM quantum computer.

TECHNOLOGIES

QUANTUM PART



Python



Qiskit



Mitiq

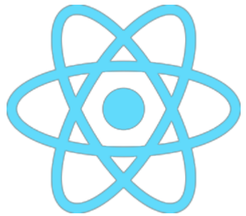
APP PART



OpenCV



FastAPI



React



GitHub



VS Code

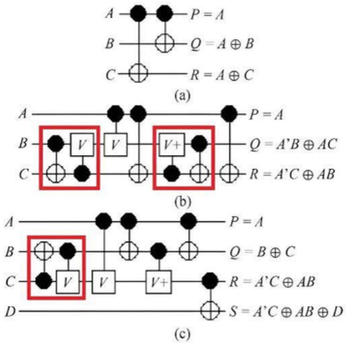
HOW OUR APP WORKS

1. Web app

User uploads a photo

2. Photo reader

Reads the numbers



3. Circuit builder

Rules → quantum circuit

4. Quantum solver

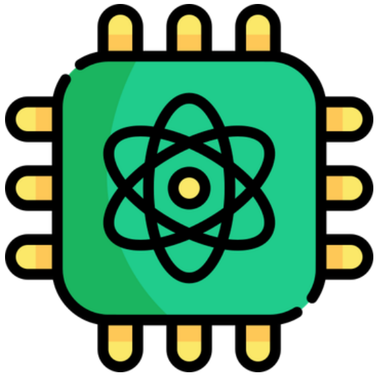
Runs Grover's search

The answer goes back to the web app (through a **FastAPI** server): the solved grid charts.

Project Plan

OUR TIMELINE

Week 1



Core Quantum Solver

Week 2



Dynamic Solver & Image Processing

Week 3



Optimization, Integration & Final Presentation and report

WHAT WE WILL DELIVER



Web app

Upload a Sudoku, see the answer and charts



Source code

Quantum solver and app on GitHub



Results

Tables: perfect vs. noisy vs. fixed



Report & demo

Final report and live presentation

QUSUDOKU IN ONE SENTENCE

A Sudoku app powered by a quantum algorithm, plus a clear test of how quantum mistakes affect it.

THANK YOU

